

Esercizio.

Let (N, E) be $\{\epsilon\}$ -structure (that is: E is the interpretation of ϵ in the structure N). Suppose $M \subseteq N$ E -transitive. Then:

1. $(M, E \cap N^2) \subseteq (N, E)$
2. Δ_0 -formulae are absolute between M and N
3. if $M, N \models ZF_{\Delta_0}^-$, then $\Delta_1(ZF_{\Delta_0})$ -formulae are absolute between M and N

Dimostrazione.

1. The first item follows directly from the definition.
2. The last item follows from the fact that a formula in Δ_1 is upward and downward absolute since it can be written as Σ_1 and Π_1 .
3. We prove the second item by induction on the complexity. As an example let's prove that $\psi(y) : \forall x \in y \phi(x)$ is absolute, where $\phi(x) \in \Delta_0$. Let $b \in M$.
 - If $N \models \psi(b)$ then for all a such that $N \models aEb$ we have $N \models \phi(a)$. Since $(M, E \cap M^2) \subseteq (N, E)$, we have that for all a such that $M \models aEb$ we have $N \models \phi(a)$. By inductive hypothesis we now get that for all a such that $M \models aEb$ we have $M \models \phi(a)$. Equivalently $M \models \forall x \in b \phi(x)$, that is $M \models \psi(b)$.
 - If $M \models \psi(b)$ then for all a such that $M \models aEb$ we have $M \models \phi(a)$. Now take $c \in N$ such that $N \models cEb$. Since $b \in M$ and M transitive, we can say $c \in M$. This means that $M \models \phi(c)$. By inductive hypothesis we have that $N \models \phi(c)$. Since c is arbitrary we have that $N \models \psi(b)$.

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